

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences across **3,900 transactions** to guide strategic decisions.



Dataset at a Glance

3,900

Transactions

18

Columns

37

Missing Values

Review Rating only

50

Locations

Features span demographics, purchase details, and shopping behavior — including discount usage, subscription status, and review ratings.

Data Preparation in Python

→ Clean & Standardize

Imputed missing Review Ratings by category median; renamed columns to snake_case

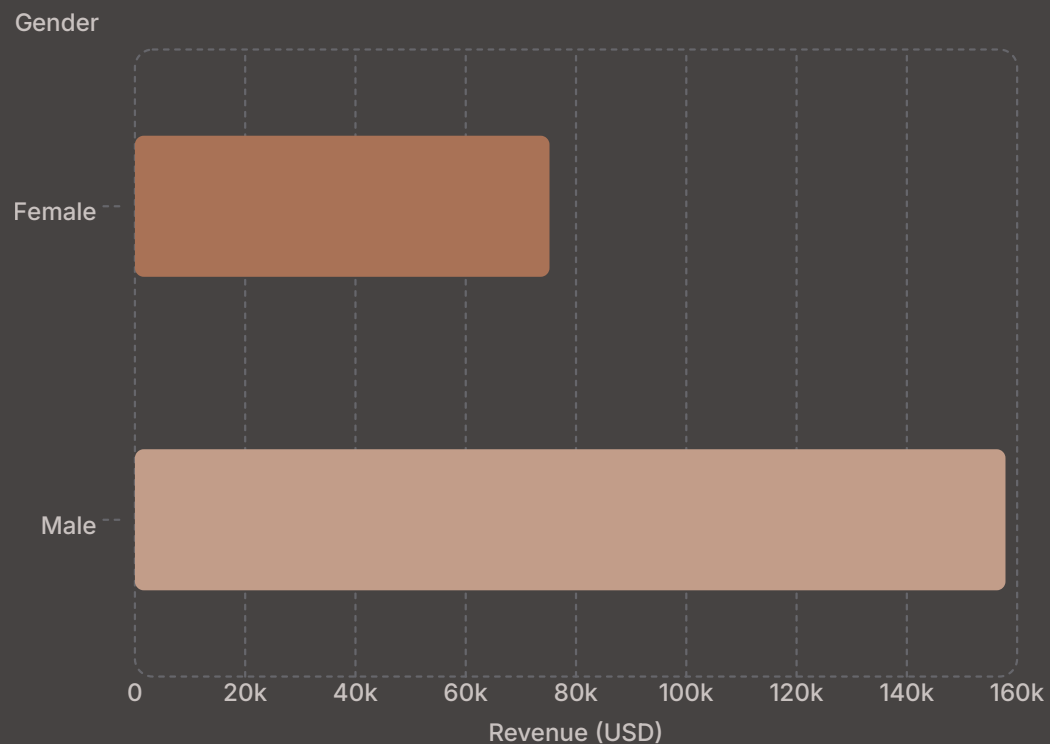
→ Feature Engineering

Created **age_group** bins and **purchase_frequency_days**; dropped redundant **promo_code_used**

→ Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis

Revenue by Gender



Key Insight

Male customers generated **2.1× more revenue** than female customers — \$157,890 vs. \$75,191.

i Male shoppers represent 68% of all transactions (2,652 of 3,900).

Products, Ratings & Shipping

Top 5 Products by Rating

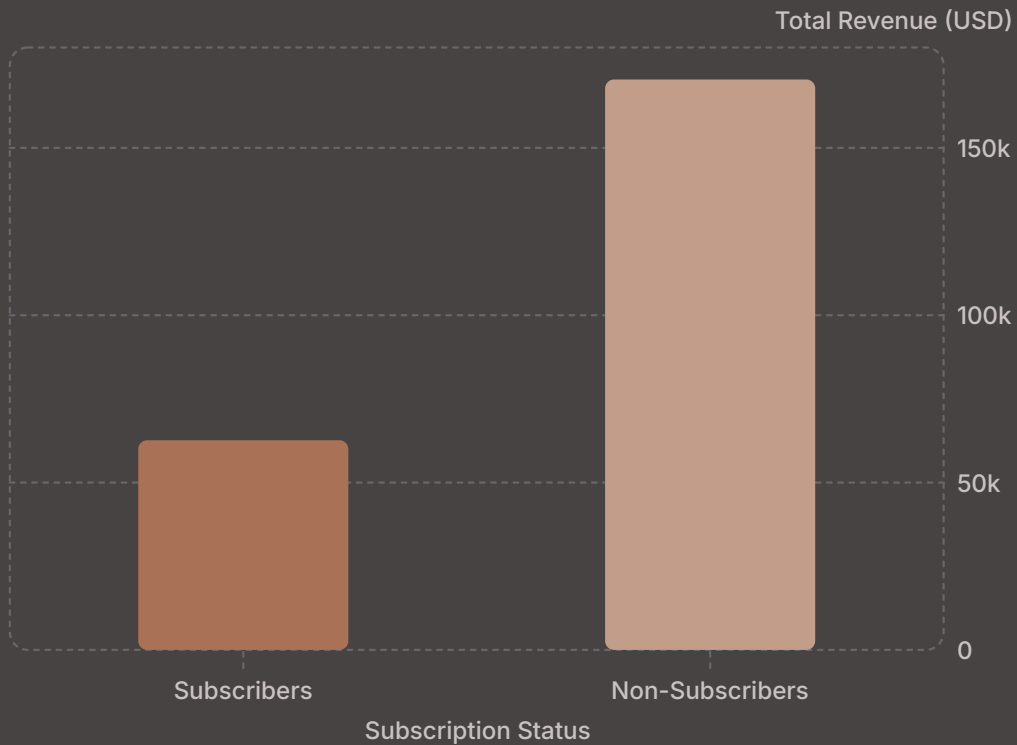
Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Shipping Type Comparison

Standard	Express
Avg spend: \$58.46	Avg spend: \$60.48

Express shippers spend slightly more — potential upsell opportunity.

Subscribers vs. Non-Subscribers



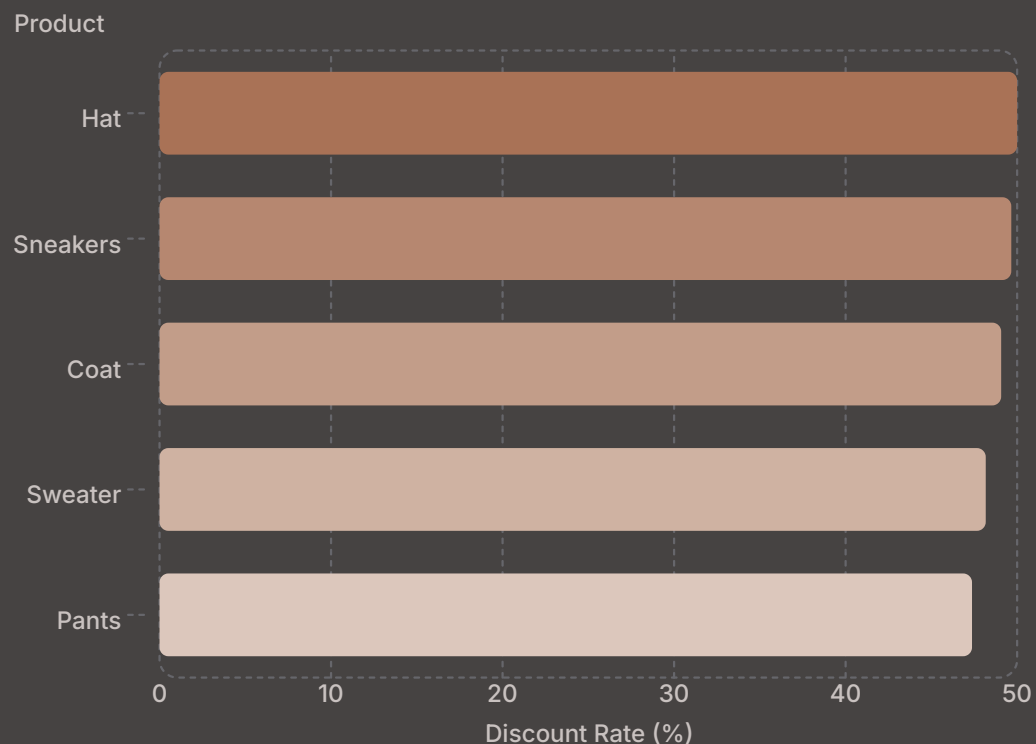
The Subscription Gap

Only **27%** of customers subscribe, yet average spend is nearly identical (~\$59.50).

📌 839 high-spending discount users identified — strong conversion targets.

Discount Dependency & Customer Segments

Most Discounted Products



Customer Segmentation

Loyal

3,116 customers

Returning

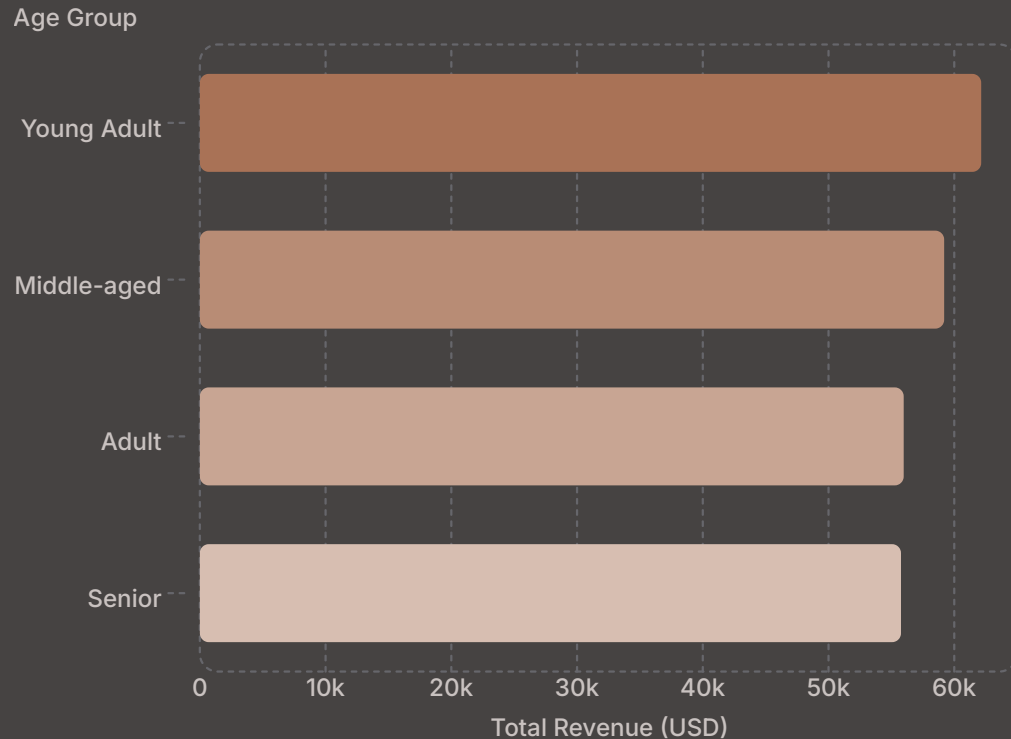
701 customers

New

83 customers

958 repeat buyers (5+ purchases) are subscribed — loyalty drives subscription.

Revenue by Age Group



Age Group Insights

Young Adults lead in revenue contribution. Revenue is relatively evenly distributed across all four age groups — each contributes ~\$55K–\$62K.

i Average customer age: **44 years** (range 18–70)

Power BI Dashboard



Interactive Filters

Subscription, Gender, Category, Shipping Type



Revenue & Sales

By category and age group



Key Metrics

3.9K customers · \$59.76 avg purchase · 3.75 avg rating



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow the Loyal segment



Review Discount Policy

Balance sales lifts with margin control



Targeted Marketing

Focus on high-revenue age groups & express shippers

